# PROWASH 365

# SOFTWARE CONFIGURATION MANAGEMENT MASTER PLAN

**SCMP**
**VERSION 1.0**

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# 1. Master Plan Overview

## 1.1 Purpose of the Master Plan

The purpose of this Software Configuration Master Plan is to establish a strategic framework for managing and controlling the configuration of ProWash365, ensuring that its evolution remains organized, traceable, and aligned with the project's operational and technical objectives. ProWash365 is a real business platform composed of a public web presence, an administrative CRM, a backend service layer, a mobile application for customers and field operations, and a database and deployment ecosystem that support cleaning-service operations.

This Master Plan does not replace the operational Software Configuration Management Plan. Instead, it coordinates the strategic direction of configuration management, baseline evolution, Software Configuration Items (SCIs), Change Requests (CRs), configuration status accounting, software auditing, release planning, and long-term architectural evolution.

- Preserve configuration integrity across code, documentation, database artifacts, deployment assets, and business modules.
- Coordinate current and future baselines from V0 to Vn.
- Provide a clear decision structure for evaluating and integrating change requests.
- Maintain traceability among baselines, CRs, SCIs, releases, repository artifacts, and audits.
- Support ProWash365 as a production-oriented business system rather than a temporary academic artifact.

## 1.2 Project Context

ProWash365 supports the digitalization of a residential and commercial cleaning service model. The platform is intended to connect customers, administrators, service packages, subscriptions, quotations, payments, employee operations, and service evidence within a controlled software ecosystem.

The repository used as a configuration reference contains the initial baseline, the SCM plan, and several SCI documents that describe controlled configuration elements for advertising, legal service and employee reporting, monitoring for city expansion, and employee mobile app photo evidence. These documents provide the historical base of the project, while this Master Plan defines the strategic control model that ProWash365 should follow going forward.

**Table 1.** Controlled source documents considered by the Master Plan.

*This table identifies the existing SCM artifacts that are coordinated by the Master Plan and explains how each one contributes to the strategic configuration framework.*

| Document / Artifact | Strategic Function | Use within this Master Plan |
|---|---|---|
| Baseline documentation | Defines the initial approved scope and reference state of the system. | Used to establish the V0/V1 reference and future baseline transitions. |
| Software Configuration Management Plan | Defines operational SCM activities, responsibilities, and procedures. | Used as the operational foundation under this Master Plan. |
| SCI-01 - Advertising Module | Controls advertisement management, segmentation, scheduling, priority, and versioning. | Used as a controlled configuration item connected to CRM and promotional business capabilities. |

| Document / Artifact | Strategic Function | Use within this Master Plan |
|---|---|---|
| SCI-02 - Legal Service and Employee Reporting | Controls legal-service reporting and employee operational reporting capabilities. | Used to extend compliance and internal operational tracking. |
| SCI-03 - Monitoring Module for City Expansion | Controls monitoring capabilities needed to expand operations to new cities. | Used to support scalability and geographic expansion. |
| SCI-04 - Employee Mobile App Photo Evidence | Controls photo-evidence capture from the employee mobile app. | Used to support service validation, customer trust, and auditability. |

### 1.3 Master Plan Scope

The scope of this document includes strategic configuration planning for all controlled ProWash365 software components, including source code repositories, backend APIs, mobile functionality, CRM functionality, data models, deployment settings, documentation, baselines, releases, and configuration audit evidence.

- Baseline evolution from V0 to Vn.
- SCI identification, ownership, versioning, dependency control, and baseline association.
- CR evaluation and integration into planned releases.
- Configuration Status Accounting (CSA) to maintain visibility of active and historical configuration states.
- Software Auditing (SA) to verify functional, physical, security, and release consistency.
- Risk management associated with uncontrolled change, deployment, data integrity, security, and scalability.

### 1.4 Governance Concept Applied to ProWash365

In this document, governance means the set of rules, roles, controls, and decision mechanisms used to manage the evolution of the ProWash365 platform. It is not a political concept. It refers to project control: who decides, what can be changed, how changes are evaluated, how versions are released, and how evidence is maintained.

## 2. Strategic Vision of ProWash365

### 2.1 Business Vision

The business vision of ProWash365 is to provide a scalable and reliable platform for managing cleaning services, customer relationships, memberships, quotations, appointments, payments, promotional content, employee execution, and service evidence. The system must support business growth without losing control over releases, requirements, data consistency, and operational traceability.

The Master Plan treats configuration management as a strategic business capability. Every relevant change must contribute to a controlled evolution of the system, preserving customer trust, operational continuity, and maintainability.

### 2.2 Operational Vision

The operational vision is centered on traceable service management. Customers should be able to register, request services, manage homes, receive notifications, and validate service execution. Administrators should be able to manage services, packages, quotes, users, promotional content, and operational records. Employees should be able to receive assignments and generate evidence of completed work when the mobile employee workflow is activated.

## 2.3 Technological Vision

The technological vision is based on a modular full-stack architecture. The expected system structure includes a web/CRM frontend, mobile applications, a backend API layer, a relational database, cloud object storage for images and service evidence, and controlled deployment environments. This architecture must evolve through baselines and releases rather than informal changes.

**Table 2.** Strategic capability alignment for ProWash365.

*This table connects the business vision of ProWash365 with the configuration management controls needed to keep each capability stable and auditable.*

| Strategic Capability | Business Objective | Configuration Management Implication |
|---|---|---|
| Customer service booking | Allow customers to request cleaning services and memberships. | Requires stable API contracts, validated data models, and controlled release testing. |
| CRM administration | Allow administrative personnel to manage users, services, quotes, packages, and content. | Requires access control, versioned UI changes, and data consistency checks. |
| Advertising and promotions | Enable targeted promotional communication. | Requires SCI control for campaigns, priority rules, dates, and segmentation. |
| Employee reporting | Support operational traceability for personnel and field activities. | Requires controlled forms, reporting workflows, and auditable records. |
| City expansion monitoring | Support growth beyond the initial operating area. | Requires scalable modules and release planning for geographic configuration. |
| Photo evidence | Support service verification and trust. | Requires controlled image storage, metadata rules, and privacy/security validation. |

## 2.4 Long-Term Evolution Strategy

The long-term strategy of ProWash365 is to evolve through controlled baselines. V0 represents the initial conceptual and documented state. V1.x represents the controlled foundation of the platform. V2.x introduces stronger operational modules and service evidence. V3.x supports expansion, scalability, and broader operational monitoring. Vn represents future controlled evolution driven by formally approved change requests.

**Table 3.** Strategic capability alignment for ProWash365.

*This table summarizes the progressive baseline strategy used to move ProWash365 from initial configuration control to future business-scale releases.*

| Evolution Stage | Main Objective | Associated Configuration Focus | Expected Baseline |
|---|---|---|---|
| Stage 0 | Establish initial documentation and configuration reference. | Initial baseline, SCM plan, repository organization. | V0 |
| Stage 1 | Stabilize core service and CRM functionality. | Core web, backend, database, CRM, service catalog. | V1.0 |

| Evolution Stage | Main Objective | Associated Configuration Focus | Expected Baseline |
|---|---|---|---|
| Stage 2 | Integrate controlled operational and promotional capabilities. | Advertising module, employee reporting, photo evidence. | V1.x |
| Stage 3 | Support geographic and operational expansion. | Monitoring module for city expansion, scalability controls. | V2.0 |
| Stage n | Continue controlled evolution through approved CRs. | Future integrations, optimizations, security upgrades, analytics. | Vn |

# 3. Organizational Control Structure

## 3.1 Control Model

The organizational control model defines how ProWash365 decisions are made and documented. The project must avoid uncontrolled modifications by using a clear structure for responsibilities, technical evaluation, customer/business validation, and release authorization.

**Table 4.** Organizational roles for ProWash365 SCM control.
*This table defines the main roles that participate in the configuration control process and clarifies what each role contributes to SCM.*

| Role | Primary Responsibility | Configuration Management Responsibility |
|---|---|---|
| Product Owner / Business Representative | Defines business priorities and validates business value. | Approves business relevance of CRs and accepts release scope. |
| Project Manager | Coordinates planning, scope, schedule, and delivery. | Ensures SCM activities are planned and executed. |
| Configuration Manager | Controls baselines, SCI registry, versioning, and status accounting. | Maintains configuration records and verifies traceability. |
| Software Architect / Technical Lead | Preserves architectural consistency and technical feasibility. | Evaluates technical impact of CRs and SCI dependencies. |
| Development Team | Implements approved changes. | Uses branches, commits, pull requests, tags, and release conventions. |
| QA / Audit Responsible | Validates functional correctness, physical consistency, and release quality. | Executes audits, test evidence reviews, and release verification. |
| Security / Infrastructure Responsible | Controls environments, deployment configuration, storage, and security rules. | Validates secrets, access, deployment, backup, and storage configuration. |

## 3.2 Configuration Control Board (CCB)

The Configuration Control Board is the decision body responsible for evaluating significant changes that affect controlled configuration items, baselines, releases, architecture, cost, schedule, or business operations. For ProWash365, the CCB may be lightweight, but it must still provide formal decision evidence.

**Table 5.** Configuration Control Board composition.

*This table presents the recommended CCB composition for ProWash365 and the type of decision input each participant provides.*

| CCB Participant | Decision Contribution | Typical Questions |
| --- | --- | --- |
| Business representative | Business value and priority. | Is this change aligned with ProWash365 operations and customer expectations? |
| Project manager | Planning, scope, and cost. | Can the change be delivered within schedule and budget constraints? |
| Technical lead | Architecture and feasibility. | Does the change affect APIs, database schema, mobile app, CRM, or integrations? |
| Configuration manager | Baseline and traceability control. | Which SCI, CR, branch, version, and release will carry this change? |
| QA / audit responsible | Validation and evidence. | What tests and audit evidence are required before integration? |

## 3.3 Decision-Making Structure

**Table 6.** Decision-making lifecycle.

*This table describes the decision process that must be followed before a significant change becomes part of a controlled ProWash365 baseline.*

| Decision Stage | Purpose | Required Output |
| --- | --- | --- |
| Request identification | Register a proposed change or configuration issue. | Change Request record or issue ticket. |
| Preliminary evaluation | Determine whether the request deserves formal analysis. | Initial acceptance, rejection, or deferral. |
| Technical analysis | Evaluate implementation effort, dependencies, architecture, and affected SCIs. | Impact analysis and affected-item list. |
| Business validation | Validate business value, cost, and operational relevance. | Priority and business decision. |
| CCB review | Approve, reject, defer, or request more information. | Formal CCB decision record. |
| Implementation planning | Assign release, branch, responsible team, and validation plan. | Implementation plan and target baseline. |
| Validation and audit | Verify the change before release integration. | Test report, audit record, and release authorization. |
| Baseline integration | Incorporate the validated change into the controlled baseline. | Updated baseline, tag, release notes, and status accounting update. |

# 4. Master Baseline Strategy

## 4.1 Baseline Governance

A baseline is a formally approved configuration state of the system. It acts as a reference point for development, testing, release, auditing, and future change control. ProWash365 uses baselines to prevent uncontrolled evolution and to make sure that each release can be traced back to approved requirements, CRs, SCIs, documentation, and repository evidence.

**Table 7.** Master baseline roadmap from V0 to Vn.

*This table defines the baseline sequence used to govern the evolution of ProWash365 from its initial controlled state to future approved versions.*

| Baseline | Configuration State | Main Scope | Control Objective |
|---|---|---|---|
| V0 | Initial documented reference. | Initial SCM documentation, baseline definition, SCI identification, repository evidence. | Establish the starting point for controlled evolution. |
| V1.0 | Core operational foundation. | Customer registration, CRM basics, service catalog, packages, backend APIs, database schema. | Create the first stable operational platform baseline. |
| V1.x | Incremental operational capabilities. | Advertising module, legal/employee reporting, improved CRM workflows, controlled mobile enhancements. | Integrate approved SCIs without destabilizing the core baseline. |
| V2.0 | Expansion and field evidence baseline. | City expansion monitoring, employee photo evidence, scalable operational processes. | Support growth and auditable service execution. |
| Vn | Future controlled evolution. | Analytics, advanced automation, payment improvements, security hardening, integrations. | Allow long-term evolution through formal CR and release control. |

## 4.2 Baseline Transition Rules

- No baseline may be changed without an approved CR or authorized maintenance record.
- Each baseline must have a clear version identifier, release notes, affected SCI list, test evidence, and approval record.
- A baseline transition must update the configuration status accounting register.
- Emergency fixes may be handled through a hotfix path, but they must be documented and merged back into the controlled baseline.
- Production releases must correspond to a repository tag or equivalent release identifier.

**Table 8.** Baseline transition control rules.
*This table identifies the conditions and evidence required to move from one ProWash365 baseline to the next.*

| From Baseline | To Baseline | Transition Trigger | Required Evidence |
|---|---|---|---|
| V-0 | V-1.0 | Completion of the first stable operational scope. | Approved baseline definition, core SCI list, repository tag, validation evidence. |
| V-1.0 | V-1.x | Approved enhancement that extends the operational foundation. | CR approval, affected SCI update, test evidence, release notes. |
| V-1.x | V-2.0 | Expansion or major operational capability integration. | CCB decision, architecture review, migration plan, audit report. |
| V-2.0 | V-n | Future strategic change or enterprise evolution. | CR portfolio, risk assessment, architecture decision record, release authorization. |

## 4.3 Baseline Traceability Model

**Table 9.** Baseline-to-CR-to-SCI traceability model.
*This table shows how baselines connect to change requests and configuration items so that future audits can reconstruct the history of the platform.*

| Baseline | Associated CRs | Associated SCIs / CIs | Strategic Focus |
|---|---|---|---|
| V0 | Initial configuration control scope. | SCM Plan, Baseline documentation, repository structure. | Initial configuration reference and project control foundation. |
| V1.0 | Core platform stabilization CRs. | Core web/CRM, backend, database, service catalog, customer app functions. | Stable operational foundation. |
| V1.x | CRs for advertising and internal reporting enhancements. | SCI-01, SCI-02, related CRM and backend components. | Business administration and operational traceability. |
| V2.0 | CRs for city expansion and employee photo evidence. | SCI-03, SCI-04, mobile app modules, storage configuration. | Scalability and auditable field execution. |
| Vn | Future approved CRs. | Future SCIs, infrastructure, analytics, integrations, security controls. | Long-term controlled platform evolution. |

# 5. Enterprise Architecture Evolution

## 5.1 Current Architecture Overview

The ProWash365 architecture is expected to be organized in layers that separate presentation, business logic, persistence, storage, infrastructure, and operational documentation. This separation allows each SCI to be assigned to concrete controlled components and reduces the risk of uncontrolled cross-impact between modules.

**Table 10.** ProWash365 enterprise architecture layers.

*This table organizes the technical structure of ProWash365 into layers that can be controlled through SCM practices.*

| Architecture Layer | Primary Responsibility | Examples of Controlled Items |
|---|---|---|
| Presentation layer | User interfaces for website, CRM, and mobile applications. | Next.js pages/components, React Native screens, forms, navigation, style configuration. |
| Service / API layer | Business logic, authentication, service management, quotes, packages, payments, reporting. | NestJS controllers, services, DTOs, guards, API contracts. |
| Persistence layer | Relational data storage and data integrity. | Prisma schema, migrations, PostgreSQL tables, indexes, relations. |
| Storage layer | Images, documents, and photo evidence. | Cloudflare R2 buckets, object keys, signed URL logic, storage policies. |
| Infrastructure layer | Deployment, environment configuration, secrets, CI/CD, hosting. | Environment variables, deployment scripts, Vercel/cloud settings, Docker files. |
| Documentation layer | Controlled requirements, baselines, SCIs, CRs, tests, audit evidence. | PDF/DOCX controlled documents, release notes, CCB records. |

## 5.2 Architecture Evolution by Baseline

**Table 11.** Architecture evolution by controlled baseline.

*This table describes how ProWash365 architecture is expected to evolve as new baselines are approved.*

| Baseline | Architecture Evolution | Main Impacted Layers |
|---|---|---|
| V-0 | Documentation and initial repository structure are controlled. | Documentation layer, repository management. |
| V-1.0 | Core CRM, backend, database, and customer-facing functions are stabilized. | Presentation, service, persistence. |
| V-1.x | Advertising, reporting, and internal management modules are incorporated. | Presentation, service, persistence, documentation. |
| V-2.0 | City expansion and photo evidence introduce scalability and storage requirements. | Mobile, service, storage, infrastructure. |

| Baseline | Architecture Evolution | Main Impacted Layers |
|---|---|---|
| V-n | Future analytics, integrations, security hardening, and automation are added. | All controlled architecture layers. |

## 5.3 Repository and Branch Strategy

Repository control must support traceability from requirements and CRs to commits, pull requests, tags, and releases. The public SCM repository provides document evidence, while the product repositories should maintain implementation evidence for web, backend, and mobile components.

**Table 12.** Repository and branch control model.

*This table defines how repository elements should be used to maintain implementation traceability for ProWash365.*

| Branch / Tag Element | Purpose | Control Rule |
|---|---|---|
| main | Stable branch representing approved production or release-ready code. | Direct commits should be restricted; changes should arrive through reviewed pull requests. |
| develop | Integration branch for approved work before release stabilization. | Must remain buildable and traceable to approved tickets or CRs. |
| feature/<cr-or-sci-id> | Branch for approved feature development. | Must reference CR or SCI identifier in branch name or pull request. |
| hotfix/<issue-id> | Branch for urgent production corrections. | Requires post-implementation documentation and baseline reconciliation. |
| vX.Y.Z tags | Release identifiers. | Each release tag must be associated with release notes, baseline status, and validation evidence. |

# 6. Configuration Management Strategy

## 6.1 Configuration Governance Principles

The configuration management strategy is based on identification, control, status accounting, verification, and auditability. Every controlled item must have an owner, version, relationship to the baseline, and a clear change path.

**Table 13.** Configuration management principles.

*This table defines the strategic SCM principles that guide all configuration activities in ProWash365.*

| Principle | Meaning for ProWash365 | Expected Evidence |
|---|---|---|
| Identification | Controlled items must be uniquely named and classified. | SCI register, CI catalog, repository paths. |
| Traceability | Changes must be linked to requirements, CRs, SCIs, commits, tests, and releases. | Traceability matrix, pull requests, release notes. |

| Principle | Meaning for ProWash365 | Expected Evidence |
|---|---|---|
| Integrity | Approved baselines must remain stable and reproducible. | Tags, deployment records, audit reports. |
| Accountability | Each configuration item must have an owner or responsible role. | Responsibility matrix, CCB records. |
| Auditability | The project must be able to prove what changed, why, when, and by whom. | CSA records, audit checklists, test evidence. |
| Standardization | Naming, versioning, documentation, and release procedures must be consistent. | Templates, naming conventions, repository standards. |

## 6.2 CI and SCI Identification

A Configuration Item (CI) is any project element that must be controlled. A Software Configuration Item (SCI) is a software-related CI or a group of related software components that require formal control. The following catalog acts as a high-level reference for ProWash365.

**Table 14.** High-level CI and SCI catalog.
*This table identifies the main controlled configuration elements and the baseline in which each item is expected to be managed.*

| ID | Configuration Item / SCI | Type | Owner | Baseline Association |
|---|---|---|---|---|
| CI-01 | SCM Plan and Master Plan documents | Documentation CI | Configuration Manager | V-0 and all future baselines. |
| CI-02 | Baseline documentation | Documentation CI | Configuration Manager | V-0, V-1.0, V-1.x, V-2.0, V-n. |
| SCI-01 | Advertising Module | Software Configuration Item | CRM / Backend responsible | V-1.x. |
| SCI-02 | Legal Service and Employee Reporting | Software Configuration Item | Operations / Backend responsible | V-1.x. |
| SCI-03 | Monitoring Module for City Expansion | Software Configuration Item | Architecture / Operations responsible | V-2.0. |
| SCI-04 | Employee Mobile App Photo Evidence | Software Configuration Item | Mobile / Storage responsible | V-2.0. |
| CI-03 | Database schema and migrations | Technical CI | Backend / Database responsible | V-1.0 onward. |
| CI-04 | Deployment and environment configuration | Infrastructure CI | Infrastructure responsible | V-1.0 onward. |
| CI-05 | Test evidence and audit records | Quality CI | QA / Audit responsible | All baselines. |

## 6.3 SCI Governance Strategy

**Table 15.** SCI governance strategy.
*This table describes the control scope, dependencies, and validation focus for each existing ProWash365 SCI.*

| SCI | Control Scope | Dependencies | Validation Focus |
|---|---|---|---|
| SCI-01 Advertising Module | Promotion content, segmentation, activation dates, priorities, CRM management. | CRM UI, backend APIs, database tables, notification/content rules. | Correct display logic, date activation, segmentation, priority resolution. |
| SCI-02 Legal Service and Employee Reporting | Operational and legal/reporting workflows linked to employees and services. | Employee data, service records, reporting APIs, permissions. | Report accuracy, access control, audit trail. |
| SCI-03 Monitoring Module for City Expansion | Operational monitoring needed to expand service control by city or region. | Location data, CRM dashboards, service availability, operations configuration. | Scalability, city-based filtering, operational consistency. |
| SCI-04 Employee Mobile App Photo Evidence | Photo capture, upload, storage, metadata, and service evidence validation. | Mobile app, backend APIs, R2/storage, service order records. | Upload reliability, metadata correctness, privacy/security controls. |

## 6.4 Versioning Strategy

**Table 16.** Versioning and identification strategy.
*This table standardizes the naming format used for releases, baselines, SCIs, CRs, and hotfixes.*

| Version Element | Format | Example | Meaning |
|---|---|---|---|
| Product release | vMajor.Minor.Patch | v1.2.0 | A deployable product release. |
| Baseline | BL-Major.Minor | v-1.0 | A formally approved configuration reference state. |
| SCI version | SCI-ID vMajor.Minor | SCI-01 v1.1 | A controlled revision of a software configuration item. |
| Change request | CR-### | CR-005 | A formally evaluated proposed modification. |
| Hotfix | vMajor.Minor.Patch-hotfix.N | v1.2.1-hotfix.1 | An urgent patch that must later be reconciled with the active baseline. |

## 6.5 Change Control Strategy

All significant changes must be documented as CRs and evaluated according to impact, urgency, cost, architectural risk, data risk, testing effort, and baseline impact. Minor changes may be handled through standard issue tracking, but they must still be linked to commits and releases.

**Table 17.** Change classification and approval strategy.
*This table classifies changes by complexity and identifies the approval and evidence required for each type.*

| Change Type | Examples | Approval Level | Required Evidence |
|---|---|---|---|
| Minor correction | Text fixes, small UI corrections, non-functional internal refactoring. | Technical lead or project manager. | Issue record, commit reference, basic validation. |
| Functional enhancement | New CRM view, new service configuration, new customer workflow. | CCB review recommended. | CR, impact analysis, test evidence, release notes. |
| Data model change | New tables, altered relations, migration scripts. | CCB approval required. | Migration plan, rollback plan, data validation evidence. |
| Security-sensitive change | Authentication, permissions, storage access, payment/security handling. | CCB and security review required. | Security review, tests, audit evidence. |
| Strategic module integration | Advertising module, city expansion, photo evidence, analytics. | Formal CCB approval required. | CR, SCI update, baseline transition plan, audit report. |

# 7. Change and Release Control

## 7.1 Change Lifecycle

**Table 18.** Change lifecycle for ProWash365.
*This table defines the path a change must follow from initial identification to baseline integration.*

| Lifecycle Step | Description | Responsible Role | Output |
|---|---|---|---|
| 1. Change identification | A need, defect, enhancement, or risk is identified. | Requester / Product Owner | Initial request or issue. |
| 2. CR registration | The request is documented as a formal CR when it affects controlled items. | Configuration Manager | CR record. |
| 3. Impact analysis | Technical, business, cost, schedule, security, and baseline impact are evaluated. | Technical Lead / Project Manager | Impact analysis. |
| 4. CCB decision | The CCB approves, rejects, defers, or requests more information. | CCB | Decision record. |
| 5. Implementation | Approved change is implemented in controlled branches. | Development Team | Code, documents, migrations, tests. |

| Lifecycle Step | Description | Responsible Role | Output |
|---|---|---|---|
| 6. Validation | Functional, integration, regression, and configuration validation are executed. | QA / Audit Responsible | Test and audit evidence. |
| 7. Release integration | The change is integrated into the target release and baseline. | Configuration Manager / Technical Lead | Release candidate and notes. |
| 8. Baseline update | The approved release becomes part of the controlled baseline. | Configuration Manager | Updated baseline and CSA register. |

## 7.2 Strategic Integration of Change Requests

**Table 19.** Strategic CR integration roadmap.

*This table aligns the expected change request categories with the SCI structure and target baselines.*

| CR Category | Strategic Objective | Associated SCI / CI | Target Baseline |
|---|---|---|---|
| CR for advertising control | Improve promotional management and targeted communication. | SCI-01 | V-1.x. |
| CR for reporting control | Improve legal, employee, and operational reporting. | SCI-02 | V-1.x. |
| CR for city expansion | Support business growth into additional cities. | SCI-03 | V-2.0. |
| CR for photo evidence | Improve service proof, customer trust, and employee accountability. | SCI-04 | V-2.0. |
| Future CRs | Support analytics, performance, payments, security, and automation. | Future SCIs / CIs | V-n. |

## 7.3 Release Planning Strategy

**Table 20.** Release planning strategy.

*This table defines the release types that ProWash365 should use, and the controls required for each release type.*

| Release Type | Purpose | Frequency / Trigger | Required Controls |
|---|---|---|---|
| Internal development release | Validate features before broader testing. | As needed during development. | Branch control, basic tests, internal notes. |
| Staging release | Validate integrated functionality in a controlled environment. | Before production release. | Regression tests, migration validation, QA checklist. |
| Production release | Deploy approved functionality to users. | After CCB/release approval. | Tag, release notes, backup/migration plan, audit evidence. |

| Release Type | Purpose | Frequency / Trigger | Required Controls |
|---|---|---|---|
| Hotfix release | Resolve urgent production defects. | Only when operational risk requires it. | Issue record, minimal patch, post-release audit, baseline reconciliation. |
| Major baseline release | Introduce a formal baseline transition. | When scope justifies baseline change. | CCB approval, baseline document update, CSA update, audit report. |

### 7.4 Release Acceptance Criteria

- All approved CRs assigned to the release are implemented or formally deferred.
- All affected SCIs are updated and reviewed.
- Database migrations have been tested and rollback considerations are documented.
- Security-sensitive settings, storage permissions, and environment variables are verified.
- Release notes, tag, test evidence, and audit records are available.
- The Configuration Manager updates the Configuration Status Accounting register.

## 8. Configuration Status Accounting

Configuration Status Accounting (CSA) is the formal recordkeeping process that maintains visibility over the current and historical state of configuration items. For ProWash365, CSA answers questions such as: What version is active? Which CRs were approved? Which SCIs are affected? Which baseline contains a feature? What evidence supports a release?

**Table 21.** Configuration Status Accounting register fields.
*This table defines the information that must be recorded to maintain visibility over the state of ProWash365 configuration elements.*

| Status Element | Information to Record | Responsible Role | Update Moment |
|---|---|---|---|
| SCI status | Identifier, owner, version, baseline, current state, dependencies. | Configuration Manager | Whenever SCI changes are approved or integrated. |
| CR status | Proposed, under analysis, approved, rejected, deferred, implemented, closed. | Configuration Manager / CCB | At every CR decision point. |
| Baseline status | Active baseline, target baseline, previous baseline, approval date. | Configuration Manager | When a baseline is created, updated, or retired. |
| Release status | Planned, candidate, approved, deployed, rolled back, closed. | Project Manager / Technical Lead | At every release stage. |
| Audit status | Pending, in progress, findings open, validated, closed. | QA / Audit Responsible | During and after audit activities. |
| Repository status | Branch, tag, commit range, pull requests, deployment reference. | Technical Lead | At release and baseline updates. |

**Table 22.** Change Request status model.
*This table standardizes the status values used to track the lifecycle of a ProWash365 Change Request.*

| CR Status | Meaning | Allowed Next Status |
| --- | --- | --- |
| Proposed | The change has been submitted but not analyzed. | Under Analysis, Rejected, Deferred. |
| Under Analysis | Impact and feasibility are being evaluated. | Approved, Rejected, Deferred, More Information Required. |
| Approved | The CCB has authorized implementation. | In Implementation, Deferred. |
| In Implementation | Development work is active. | In Validation, Blocked. |
| In Validation | The change is being tested or audited. | Implemented, Rework Required. |
| Implemented | The change is integrated into a release candidate or baseline. | Closed, Reopened. |
| Closed | All implementation, validation, and documentation evidence is complete. | Reopened only by formal decision. |

**Table 23.** SCI status model.
*This table defines the standard states used to manage Software Configuration Items throughout their lifecycle.*

| SCI Status | Meaning | Control Action |
| --- | --- | --- |
| Proposed | A new SCI has been identified but not yet approved. | Evaluate scope, owner, dependencies, and baseline association. |
| Approved | The SCI has been accepted as a controlled item. | Create or update the SCI document and assign version. |
| In Development | Implementation or modification is active. | Track branch, commits, tests, and affected components. |
| Under Validation | The SCI is being tested or audited. | Execute functional, physical, security, or release audits as needed. |
| Integrated | The SCI version is part of an approved baseline or release. | Update CSA register and release notes. |
| Retired | The SCI is no longer active. | Preserve historical traceability and retirement decision. |

# 9. Software Auditing Strategy

Software Auditing (SA) verifies that the software, documentation, configuration records, and releases correspond to approved requirements, CRs, SCIs, and baselines. Auditing is required before major baseline transitions and recommended before every production release.

**Table 24.** Software audit categories.
*This table defines the software audits required to protect configuration integrity throughout the ProWash365 lifecycle.*

| Audit Type | Objective | Evidence Reviewed | When Applied |
|---|---|---|---|
| Functional Configuration Audit (FCA) | Verify that implemented functionality satisfies approved requirements and CRs. | Requirements, CRs, test cases, acceptance evidence. | Before release approval or baseline transition. |
| Physical Configuration Audit (PCA) | Verify that the actual repository, files, documents, and release package match the approved configuration. | Repository tags, files, documents, deployment artifacts. | Before baseline approval and after major release packaging. |
| Release Audit | Verify that the release is complete, tested, documented, and traceable. | Release notes, version tag, test report, deployment checklist. | Before production deployment. |
| Security Audit | Verify security-sensitive settings and controlled access. | Authentication changes, environment variables, storage permissions, secrets handling. | Before security-sensitive release or infrastructure change. |
| Traceability Audit | Verify that CRs, SCIs, commits, tests, and baselines are linked. | Traceability matrix, CSA register, pull requests, audit logs. | Periodically and before baseline transitions. |

**Table 25.** Software audit checklist summary.

*This table provides a concise audit checklist that can be used before release or baseline approval.*

| Audit Checkpoint | Verification Question | Pass Criteria |
|---|---|---|
| SCI documentation | Does each affected SCI have an updated version and owner? | SCI document exists and matches implemented scope. |
| CR traceability | Can every release change be traced to an approved CR or issue? | All changes have valid references. |
| Repository consistency | Does the release tag match the approved source code state? | Tag, commit range, and release notes are consistent. |
| Database consistency | Are migrations controlled, tested, and reversible when possible? | Migration plan and validation evidence exist. |
| Deployment consistency | Are environment variables, storage rules, and deployment settings verified? | Deployment checklist is complete. |
| Testing evidence | Are tests and validation results available? | Functional/regression tests pass or exceptions are approved. |

# 10. Operational SCM Plan

The operational SCM Plan defines the recurring activities required to keep the Master Plan active. It translates strategic configuration management into scheduled and repeatable work.

**Table 26.** Operational SCM activity plan.

*This table defines the recurring SCM activities that must be performed to keep ProWash365 under controlled configuration management.*

| SCM Activity | Description | Frequency / Trigger | Responsible Role |
| --- | --- | --- | --- |
| SCI register maintenance | Update identifiers, owners, versions, dependencies, and statuses. | Whenever SCI changes occur; review monthly. | Configuration Manager. |
| CR review meeting | Review proposed and active change requests. | Weekly or when significant CRs are submitted. | CCB / Project Manager. |
| Repository review | Verify branches, pull requests, tags, and release references. | Before every release and monthly. | Technical Lead. |
| Baseline review | Verify whether current scope requires a new baseline or baseline update. | At release milestones or major CR approval. | Configuration Manager / CCB. |
| Status accounting update | Update CR, SCI, baseline, audit, and release status records. | At each change lifecycle transition. | Configuration Manager. |
| Pre-release audit | Verify release completeness, traceability, and quality evidence. | Before staging or production release. | QA / Audit Responsible. |
| Post-release review | Review deployment results, incidents, rollback needs, and documentation updates. | After each production release. | Project Manager / Technical Lead. |

**Table 27.** SCM deliverable control plan.

*This table defines the main SCM deliverables and the minimum information that each one should contain.*

| SCM Deliverable | Purpose | Minimum Content | Storage / Control Location |
| --- | --- | --- | --- |
| Master Plan | Strategic configuration control document. | Vision, baselines, strategy, CSA, SA, risk, release model. | Controlled document repository. |
| SCM Plan | Operational configuration management procedures. | Roles, procedures, tools, workflows, audit process. | Controlled document repository. |
| SCI document | Detailed description of each controlled item. | Scope, version, owner, dependencies, status, validation. | Repository / documentation folder. |
| CR document | Formal change request and decision record. | Problem, proposal, impact, cost, risk, CCB decision. | Change control folder or issue tracker. |
| Baseline record | Formal approved configuration state. | Version, included SCIs, release tag, approvals, evidence. | Baseline folder and CSA register. |
| Audit report | Evidence of functional, physical, security, or release verification. | Checklist, findings, decisions, corrective actions. | Audit evidence folder. |
| Release notes | Summary of released changes. | Version, features, fixes, migrations, known issues. | Release page / repository tag. |

**Table 28.** RACI matrix for SCM responsibilities.

*This table summarizes responsibilities using R = Responsible, A = Accountable, C = Consulted, and I = Informed.*

| Responsibility | Product Owner | Project Manager | Configuration Manager | Technical Lead | Development Team | QA / Audit |
|---|---|---|---|---|---|---|
| Approve business priority | A | C | I | C | I | I |
| Register CR | C | C | R/A | C | I | I |
| Analyze technical impact | I | C | C | R/A | C | C |
| Approve baseline transition | A | A | R | C | I | C |
| Implement approved change | I | I | I | A | R | C |
| Validate release | C | C | I | C | C | R/A |
| Update CSA register | I | C | R/A | C | I | C |

# 11. Risk Management Strategy

Risk management within the Master Plan focuses on preventing uncontrolled configuration growth, deployment instability, loss of traceability, security weaknesses, and operational interruptions. Risks should be reviewed during CR evaluation, release planning, and baseline transitions.

**Table 29.** Risk management matrix.

*This table identifies major ProWash365 configuration risks and the mitigation strategies that should be applied during project evolution.*

| Risk Category | Specific Risk | Potential Impact | Mitigation Strategy |
|---|---|---|---|
| Configuration risk | Uncontrolled changes bypass CR or review. | Loss of traceability and unstable releases. | Use CCB approval, branch protection, pull requests, and CSA updates. |
| Baseline risk | Active baseline does not match repository or deployed version. | Audit failure and operational confusion. | Use release tags, baseline records, and physical configuration audits. |
| Data risk | Database migration breaks existing customer, service, or order data. | Operational interruption and data inconsistency. | Require migration testing, backups, rollback plan, and schema review. |
| Security risk | Unauthorized access to APIs, storage, or administrative functions. | Information exposure or misuse. | Security reviews, access control validation, secrets management, storage permission audits. |

| Risk Category | Specific Risk | Potential Impact | Mitigation Strategy |
|---|---|---|---|
| Operational risk | Feature release affects bookings, quotes, packages, or service records. | Customer and administrative disruption. | Use staging validation, regression tests, release windows, and rollback planning. |
| Scalability risk | City expansion increases workload beyond current architecture. | Performance degradation and operational bottlenecks. | Plan BL-2.0 scalability controls, monitoring, and infrastructure review. |
| Documentation risk | SCI, CR, and baseline documents become outdated. | Poor auditability and unclear decisions. | Make documentation updates part of release acceptance criteria. |

**Table 30.** Risk level response model.

*This table defines how risk severity determines the level of control and approval required.*

| Risk Level | Criteria | Required Response |
|---|---|---|
| Low | Limited impact, no baseline change, low implementation complexity. | Track in issue system and validate through normal review. |
| Medium | Affects controlled functionality, requires testing, limited baseline impact. | Register CR, analyze impact, and include in release plan. |
| High | Affects data, security, architecture, payment, storage, or production stability. | Formal CCB review, audit plan, rollback strategy, and release approval. |
| Critical | May interrupt service, compromise information, or invalidate baseline integrity. | Executive/business escalation, emergency plan, hotfix process, post-incident audit. |

# 12. Conclusions

This Software Configuration Master Plan establishes the strategic structure required to guide the controlled evolution of ProWash365 as a real business platform. The document coordinates existing SCM artifacts, baseline history, SCI definitions, change control, release planning, status accounting, software auditing, and risk management into one integrated control framework.

The Master Plan positions configuration management as an operational and strategic discipline. Its purpose is not only to document what has already been created, but to ensure that future development remains organized, traceable, auditable, and aligned with the business needs of ProWash365.

By applying baseline control from V0 to Vn, maintaining a structured CI/SCI catalog, formally evaluating CRs, updating configuration status accounting records, and executing software audits before major releases, ProWash365 can continue evolving without losing configuration integrity or operational stability.

# Appendix A. Repository and Controlled Document References

The following public repository was used as a reference point for the SCM documentation history and controlled artifacts associated with ProWash365: https://github.com/dazidev/SOFTWARE-CONFIGURATION-MANAGEMENT

**Table 31.** Repository reference summary.

*This table summarizes how the repository and its controlled documents support the Master Plan.*

| Reference | Description | SCM Use |
| --- | --- | --- |
| Public SCM repository | Repository containing baseline, SCM plan, and SCI documentation. | Source of controlled documentation evidence. |
| Baseline document | Defines the initial configuration baseline and reference state. | Used to structure V0/V1 baseline control. |
| Plan for Designing an SCM V1 | Operational SCM plan document. | Used as the operational management layer under this Master Plan. |
| SCI V1 / SCI V2 and SCI-02 to SCI-04 documents | Software Configuration Item documentation. | Used to define controlled software elements and future baseline integration. |